Monte Carlo Simulation of Light Propagation in Skin Tissue Phantoms Using a Parallel Computing Method

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ABSTRACT

In Monte Carlo simulations of light propagating in biological tissues, photons propagating in the media are described as classic particles being scattered and absorbed randomly in the media, and their path are tracked individually. To obtain any statistically significant results, however, a large number of photons is needed in the simulations and the calculations are time consuming and sometime impossible with existing computing resource, especially when considering the inhomogeneous boundary conditions. To overcome this difficulty, we have implemented a parallel computing technique into our Monte Carlo simulations. And this movement is well justified due to the nature of the Monte Carlo simulation. Utilizing the PVM (Parallel Virtual Machine, a parallel computing software package), parallel codes in both C and Fortran have been developed on the massive parallel computer of Cray T3E and a local PC-network running Unix/Sun Solaris. Our results show that parallel computing can significantly reduce the running time and make efficient usage of low cost personal computers. In this report, we present a numerical study of light propagation in a slab phantom of skin tissue using the parallel computing technique.

Keywords: Tissue scattering, Monte Carlo simulation, parallel computing

1. INTRODUCTION

The scattering of light by turbid media has been studied extensively in the past.[1] However, the existing theoretical models are still not satisfactory for explaining the experimental data in many important applications related to the light propagation in highly scattering turbid media such as the biological materials. Understanding the interaction between light and biological materials is critical in development of new optical methods for biomedical imaging applications. For this purpose, it is essential to develop efficient modeling tools in the investigation of interaction between laser radiation and turbid media.

In a turbid medium, light is scattered and absorbed due to the inhomogeneities and absorption characteristics of the medium. When the medium becomes highly scattering, multiple scattering effects become dominant, and one widely used approach to solve this type of problem is the radiative transport theory [1] which concerns transport of light energy. Within the framework of radiative transfer theory, light propagating in a turbid medium is treated as a large number of photons, with no phase and polarization characterizations, that undergo random scattering and absorption. However, the radiative transfer equation can not be solved analytically without approximations except for a few cases with simple boundary conditions. Several approximations have been developed. Among them are the first-order solution,[2] the discrete ordinates method, [2] the Kubelka-Munk two-flux and four-flux theory, [3] and the diffusion theory.[4, 5] These methods all have their limits.

Light transportation in turbid medium can be statistically simulated by various Monte Carlo methods.[6,7] Using a simple model of random walk, a Mote Carlo method can be applied to solve radiative transfer problems accurately with virtually any boundary conditions. However, the heavy demand of computation time of the Monte Carlo simulation to reduce statistical error has impeded the wide applications of the method. Recently, we have developed a new Monte Carlo method through a "time slicing" algorithm to directly calculate light distribution in turbid medium such as biological tissues.[8-10] This method is capable of direct calculation of the light distribution with inhomogeneous boundary conditions. Two examples of the inhomogeneous boundary conditions are: a converging laser beam incident on a tissue phantom with a plane surface in which the incident angle varies with the photon location, and a tissue phantom with rough interfaces in which the incident angle

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varies with fluctuating direction of the surface normal even for a collimated beam. To take the advantage of the rapid progress in personal computers (PC) with ever increasing computation power and decreasing cost, we have established a PC cluster for parallel computing to fully appreciate the potentials of the recently developed Monte Carlo method within a modest research budget.

In this paper we report our initial results in carrying out efficient Monte Carlo simulations on light distribution in biological tissue phantom by implementing parallel computing techniques, and we show that it can dramatically reduce the program running time. The material of the paper is organized as following. After a brief summary of the Monte Carlo simulation in tissue scattering in next section, the implementation of the parallel computing is then discussed, and the results of Monte Carlo simulations of a converging laser beam propagating through a tissue slab using the parallel method is then presented afterward.

2. MONTE CARLO SIMULATION OF TISSUE SCATTERING

In this section we will give a brief summary of a recently developed Monte Carlo simulation method in study the scattering and propagation of a converging laser beam in highly scattering medium.[8-10] In the present case the medium is a slab, as shown in Figure 1.

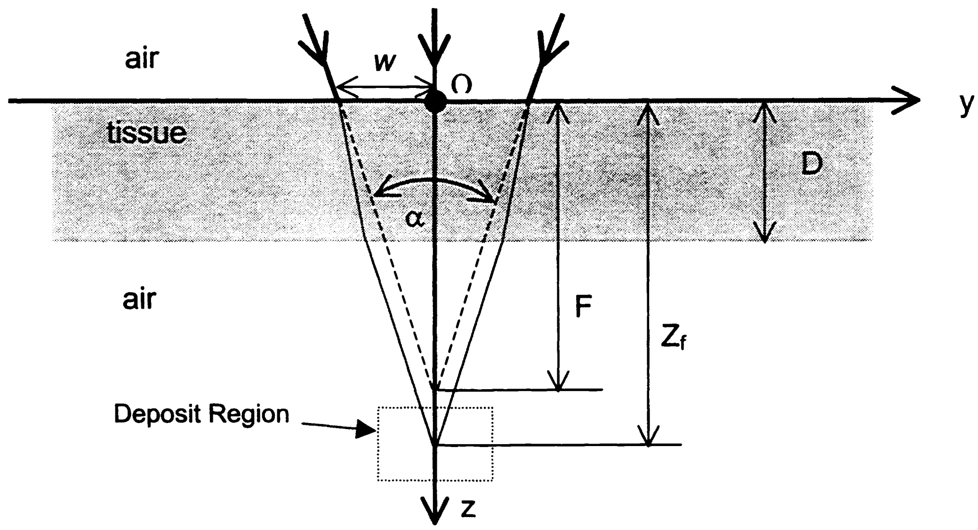

Fig. 1 Schematic of a focused laser beam propagating through a tissue slab. Where α is the cone angle and w is the radius of the beam at the entrance surface. The tissue slab has a thickness of D and an index of refraction of n. The dashed line indicates the focal point in the absence of the tissue slab that is located at a distance F below the entrance surface. In the presence of the slab, the beam will be focused in a spreading line along z-axis centered at a distance Z_f below the entrance surface.

The converging beam is incident on the tissue phantom from the air, it has a cone angle of α and a radius of w at the air-tissue interface, and its focal point in the absence of the tissue slab is located at a distance F below the exit surface. In the presence of the slab, the beam will be focused in a spreading line along z-axis centered at a distance Z_f below the entrance surface. A Cartesian coordinate system is used for the simulation with the origin set on the beam axis at the entrance surface or the xy-plane. The tissue slab is assumed to be macroscopically homogeneous with a thickness of D. It is optically characterized by an index of refraction n, scattering coefficient μ_s, absorption coefficient μ_a, and anisotropy factor g. The propagation of a photon in the phantom is described by its position and propagating direction, where the position is described by the Cartesian coordinates, and propagation direction is described by a set of moving spherical coordinates (ϕ, ψ) attached to the photon. The boundary condition at the $z = 0$ plane for each photon contained in the beam are decided by its position at the entrance surface (x_0, y_0) and its incident angle. The incident angle of each photon is to be calculated according to its distance $\rho_0 =$

$\sqrt{x_0^2 + y_0^2}$ from the z-axis and the distance between the focal point of the beam in the absence of the slab and the entrance surface, F. At the entrance surface, a photon will either be refracted or reflected according to a probability decided by the Fresnel reflectivity at the photon's incident angle. The reflected photons are not considered further.

If a photon pass through the entrance surface, it starts to propagate inside the material in the direction of the refraction angle until scattered or absorbed. When a scattering occur, the scattering angle, ϕ_s, i.e. the angle between the propagation directions before and after scattering, is randomly chosen from a distribution governed by the Henyey-Greenstein phase function.[11] The azimuthal angle, ψ_s, is randomly chosen to determine the projection of the new direction of the scattered photon in the plane perpendicular to the original one. Both of the angles can be found from the following equation: [7]

$$\varphi_s = \cos^{-1}\left(\frac{1}{2g}\left[1 + g^2 - \left(\frac{1-g^2}{1-g+2gRND}\right)^2\right]\right) \qquad (1)$$

$$\psi_s = 2\pi RND$$

where RND is a random number ranging from 0 to 1. If the photon direction before scattering is given by (ϕ,ψ), the photon direction after scattering, (ϕ',ψ'), can be related to (ϕ,ψ) and (ϕ_s, ψ_s) as [7]

$$\varphi' = \cos^{-1}(\cos\varphi_s \cos\varphi + \sin\varphi_s \sin\varphi \cos\psi_s), \qquad (2)$$

$$\psi' = \begin{cases} \psi + \tan^{-1}(\sin\varphi_s \sin\psi_s / \alpha), & for \quad \alpha > 0 \\ \psi + \tan^{-1}(\sin\varphi_s \sin\psi_s / \alpha) + \pi, & for \quad \alpha < 0 \end{cases}, \qquad (3)$$

$$\alpha = \cos\varphi_s \sin\varphi - \sin\varphi_s \cos\psi_s \cos\varphi. \qquad (4)$$

The distance traveled by a photon between successive scattering events, L_s, is randomly chosen from an exponential distribution function and given by $L_s = -\ln(1 - RND)/\mu_s$ with a mean value $< L_s >= 1/\mu_s$.[7] If a photon travels in a direction (ϕ,ψ) after a scattering event at (x, y, z) and the next scattering occurs at point (x', y', z') of a distance L_s away, the coordinates of these two points are related through the following relations

$$\begin{aligned} x' &= x + L_s \sin\varphi \cos\psi \\ x' &= y + L_s \sin\varphi \sin\psi \,\cdot \\ z' &= z + L_s \cos\varphi \end{aligned} \qquad (5)$$

As for the photon absorption, we used an approach different from previously published ones [6,7] for its offering a clear and intuitive way to the direct calculation of light distribution. For any photon which passed through the entrance surface, a life-time traveled distance in the medium, L_a, is first determined to predetermine the distance the photon may travel in the medium before it is absorbed. For an arbitrary photon, L_a is randomly chose according to an exponential distribution function and given by $L_a = -\ln(1 - RND)/\mu_a$ with a mean $< L_a > = 1/\mu_a$[7].

In this paper we are interested in the distribution of the transmitted light near the geometric focal point at z = Z_f of the refracted incident beam for a cw incident beam. To obtain the light distribution, a cubic region surrounding the focal point (which is indicated by the deposit region in Fig.1) is selected and divided into cubic grid cells, and each cell has a register which will count the number of photons falling in the cell. According to a "time-slice" method,[8] the total number of photons falling into one cell from an impulse beam can be used to calculate the steady-state number of photons in that cell for a cw beam. Thus the registers of the cells will provide the photon density distribution in the deposit region.

We track each photon along its 3-d trajectory and record its total traveled distance, L, in the medium at each scattering event. Before the photon is allowed to propagate further, L is compared with the predetermined L_a. If $L > L_a$, the photon is then eliminated as a result of absorption. Otherwise the photon's position is further checked to determine if it is on a boundary of the considered region in the turbid material. When the photon is on the air-phantom boundary, it will be either reflected back into the material, with a probability equal to the Fresnel reflection coefficient, or refracted into the air. If it is refracted into the free-space region above the tissue, it will not be tracked further. If it is refracted into the free-space region below the slab, i.e. the transmission region, it will be further checked to see if it falls into the deposit region where its presence will be recorded by the registers in each cell as it passes through. The photon will also be eliminated if it reaches other borders of the considered region. If the photon survives these tests it will be allowed to propagate further until one of the eliminating conditions is met. The procedures are repeated to the next photon until all the photons contained in the beam are depleted.

3. PARALLEL COMPUTING

The Monte Carlo simulations described above offers a flexible yet rigorous approach toward modeling of light transportation inside turbid media within the framework of the radiative transfer theory. Furthermore, its capability of handling of difficult boundary conditions related to the propagation of converging laser beam and rough interfaces has provided an extremely useful tool in understanding the interplay between the coherent and the diffuse components of the light transmitted through biological tissues.[9,10] However, this method requires large-scale computation to achieve satisfactory accuracy for its statistical nature and thus greatly needs the adoption of high performance computing techniques. But on the other hand, the tracking of individual photons are independent processes because of the nature of the radiative transfer framework. These operational features of the Monte Carlo simulations of light transportation in turbid media make itself a very good candidate for parallel computing.

To adopt parallel computing techniques for our Monte Carlo simulations, two parallel computing models can be used: either on a massively parallel processing architecture computer or a cluster of PCs. With the wide availability of parallel computing software packages and high power PCs with low cost, the latter is more attractive. Among the current parallel computing interface models we selected the message passing model over the share memory model for the its suitability to heterogeneous networks of workstations or PCs. The message passing is a paradigm used widely on parallel computers, especially scalable parallel computers with distributed memory. While several implementation methods exist for the message passing model, two stand out because of their efficiency, portability, and easy to use. These are the PVM – the Parallel Virtual Machine and MPI – the message Passing Interface. We have successfully converted our sequential program into parallel code using PVM, and our discussions and results presented here are associated with PVM, and the effort to use MPI is currently in progress.

PVM: PVM is a free software package that can be downloaded from Oak Ridge National Lab web site. It allows a programmer to create and access a concurrent computer system made from networks of loosely coupled processing elements. This ability to bring together diverse computer architectures under a central control enables the PVM users to divide a simulation problem into subtasks and assign each subtask to be executed on the processor architecture that is best suited for the subtask. PVM allows messages passing between machines that do not share data representations.

Workload distribution: To perform parallel processing, it is necessary to distribute the workload across the networked processing elements (PE). There are two main programming approaches that codes are written under PVM: master/worker and hostless. We choose the former approach to conduct our parallel processing. Under the master/worker paradigm, one task is used as the master. The sole purpose of the master task is to create all other tasks for solving the problem, to coordinate the input of initial data to each task and collect the output of results from each task. In our case, the photons contained in the incident beam are divided into multiple groups by the master program and each group is sent to a task that may run on a different PE. In each task, after receiving its group of the incident photons from the master program, the trajectories of each photon will be tracked, and any deposit into the deposit area is recorded and will be sent back to the master program when the task finishes its tracking procedure. The master will append the deposit results to the existing data from finished tasks, it will also send the PE another task to be executed until all simulations finish. Then the master program will conduct the final summarization of the deposit from all the tasks, and carry out certain required statistical calculations before terminating the program.

Cray T3E: We have conducted part of Monte Carlo simulations reported here on the massive parallel computer Cray T3E at North Carolina Super Computer Center (NCSC) where PVM is readily available and easy to use.

PC cluster: To make our parallel codes more accessible for regular users, we have established a local network of PC's to execute bulk of the simulation jobs. The PC cluster consists of fifteen Pentium PCs with Celeron CPU of 433, 466 and 500MHz and each PC has 128MB memory. The local network is served with a 100 Mbit/s network hub and the Solaris 7 operating system (the UNIX operating system for Intel x86 processors) from the Sun Microsystem. PVM is used on this network to implement the parallel process using each PC as a PE. With this local network we were able to significantly raise the simulation speed and reduce CPU time, and to perform large scale code testing and transfer large part of our numerical simulations from the CRAY supercomputers at NCSC to this dedicated PC cluster with a very modest cost.

Random number generators (RNG): One of the major challenge in our implementation of parallel computing for the Monte Carlo simulations is to find an appropriate RNG. As discussed in the previous section, calculating light distribution in the tissue phantom relies on photon tracking which contains many random processes. The scattering frequency, the scattering angle and the life-time travel length as well as the reflection at the interfaces are randomly chosen from corresponding distributions, and a good RNG is essential for obtaining accurate statistical results. Beside the standard requirements for a good RNG, an appropriate RNG needed in our simulation must be able to generate a sequence of random numbers satisfying statistical tests for randomness, uniformly distributed in the full range from 0 to 1, not correlated, and have long period within the acceptable error ranges. As we move from sequential to parallel computing, additional requirements arise that are especially important for executing parallel computing:

- The sequences of random numbers generated on each PE should satisfy the requirements of a good sequential generator.
- There should be no correlation between the sequence on different PE.
- It should generate same sequence for different PEs.

In the early days of parallel computing, two techniques were often used to derive parallel RNG from serial RNG: sequence splitting and the leapfrog method. Both of them have been reported to have their problems [12, 13]. To avoid the problems associated with both of the above approaches, recent research has lead to development of parallel random number generators based on parameterization.[14]

For our initial calculations, we employ the sequence splitting method to generate the random numbers from a tested sequential generator. We designate a task to control the distribution of random numbers among the tasks of tracking different photons. The control task initiates a block of seeds for master program for each task it spawn, and it will supply more for the same task if needed. This method does lead to a problem, though, in which it is impossible to pre-determinate accurately the number of random numbers a photon tracking task needs during the process. So the block of seeds passed to a task may not be all used up by it. Because of this problem, we found that the results of simulating a converging laser beam varies slightly if the task number differs. Results with different optical parameters shown in this paper are obtained using the same number of task to maintain the consistency and it is to demonstrate the feasibility of the parallel calculation light distribution in tissue phantom. Currently, efforts to make use of a package using the parameterization method, the Scalable Parallel Random Number Generator (SPRNG),[15] is underway.

4. RESULTS

In this section we present our results on Monte Carlo simulation of tissue scattering using the parallel computing method described in the previous sections. Here we consider a converging laser beam, with a Gaussian profile at the entrance surface, propagating through a slab of tissue phantom made of intralipid solution, as shown in Fig. 1.

To understand the results better, let us first exam the components of the light propagating inside a turbid medium. When a beam of light is incident on a turbid medium, most of the photons will be scattered before they emerge from the medium and thus form the scattered part of the transmitted light. And a very small portion of the photons will come out without scattering (the unattenuated part), if the optical pathlengths involved are not too large. Within the scattered part, some photons are scattered in the forward direction for only a few times and was named quasi-ballistic or the snake component, the rest forms the diffuse component since their distribution can be well described by a diffusion equation.[2, 16] Both unattenuated and quasi-ballistic photons have their trajectories either unchanged or slightly modified from the original. In a configuration of a converging beam propagating through a slab of turbid medium, the ballistic component will be brought together near the focal point to form a peak.[8] The quasi-ballistic component will mostly provide a photon density near the focal point and thus contribute to the background near the peak and decrease quickly away from the focal point. For the diffuse component, the photons which have been multiply scattered, will form a rather uniform background in the focal plane. If we name the radiance inside the medium as $I(\vec{r},\hat{s})$, it can be divided into the coherent part I_c and the scattered part I_d: $I(\vec{r},\hat{s}) = I_c(\vec{r},\hat{s}) + I_d(\vec{r},\hat{s})$.[2] The coherent radiance, i.e., the unattenuated part, is reduced according to:

$$\frac{dI_c(\vec{r},\hat{s})}{ds} = -\mu_t I_c(\vec{r},\hat{s}), \tag{6}$$

where μ_t is the attenuation coefficient defined as $\mu_t = \mu_s + \mu_a$. The unattenuated photons arriving near the focal point in our simulation can then be compared to that expected from Eq. (6). For a cw converging Gaussian beam with N_0 photons per unit time transmitted through a tissue slab of thickness D, according to Eq. (6) the number of unttenuated photons in the transmitted light arriving at the focal point is given by:

$$N_{uatt} \simeq (1-R_1)(1-R_2)N_0 e^{-\mu_t D} - N_0 \frac{\mu_t D w}{\sqrt{2}F}\int_{t_0}^{\infty} e^{-t^2}\,dt, \tag{7}$$

where $t_0 = \sqrt{2}F / w + \mu_t Dw / 2F$, and R_1, R_2 are the reflectivity at the entrance and exit surfaces of the slab, respectively. For the cases we considered in this paper with $F = 63mm$ and $w = 4.86mm$, we find $F >> w$, and therefore the second term in Eq. (7) can be neglected. So we have

$$N_{unatt} \simeq (1 - R_1)(1 - R_2) N_0 e^{-\mu_t D} . \quad (8)$$

This provides the expected photon density peak at the focal point. Eq. (8) also indicates that the coherent part of the incident radiance is attenuated exponentially with the optical depth μ_t D. Thus, measuring of this peak against the slab thickness can lead to the determination of the attenuation coefficient μ_t, or *vice versa*.

To study such dependence of the peak height on μ_t, we carried out simulations of a converging laser beam propagating through a slab of diluted intralipid solution using the parallel technique. Through the simulations, the slab thickness is fixed at $D = 17mm$ while μ_t is changed from 0.546 (mm^{-1}) to 1.092 (mm^{-1}) through the variation of the concentration of the solution slab. The optical parameters of intralipid solution are given by an albedo of μ_s/μ_a = 0.9666, and a refractive index $n=1.33$ and $g=0.48$.[17] We chose the transverse dimension of the slab in the *xy*-plane to be of $50mm \times 50mm$ to accommodate the simulation, and the parameters of the incident Gaussian beam to have radius w = 4.86mm, cone angle α = 8.82° and focal length F = 63mm. The unattenuated photons distribute along a spreading line in the z-axis centered at z = Z_f = 67.3mm, which is referred to as the focal point. The grid cells of the deposit region are cubic with 50μm sides and the region occupies a (16mm)^3 cube centered at the focal spot.

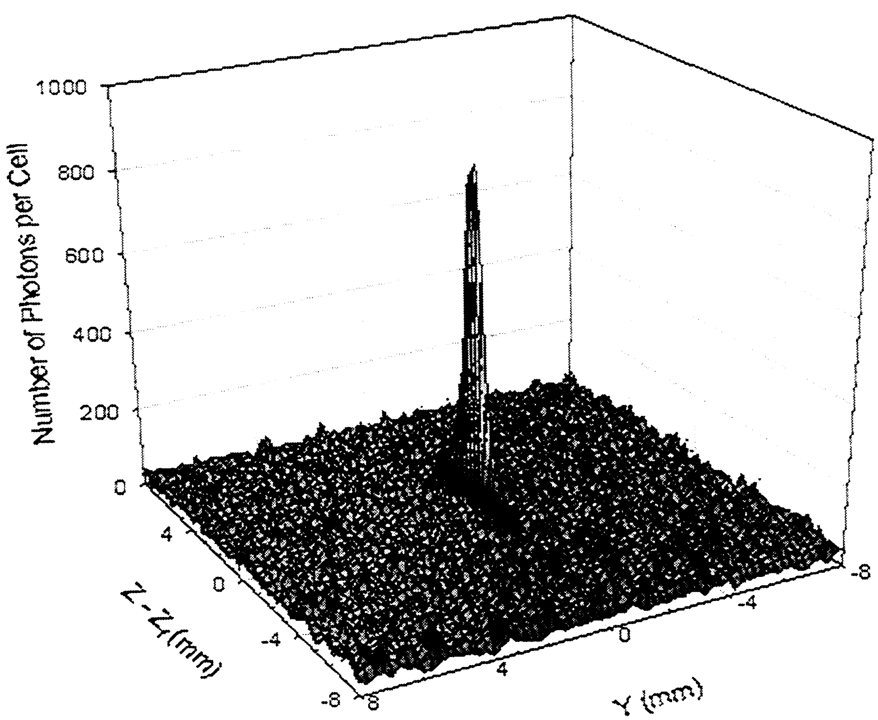

Fig. 2 Parallel computing results of Monte Carlo simulation of the photon density distribution in *yz*-plane near the focal point for a converging laser beam transmitting through a tissue phantom slab with μ_t = 0.936(mm^{-1}). The peak formed by unattenuated photons is located at the focal point z = Z_f = 67.3mm and y = 0.

Results for all the values of μ_t up to 1.09 (mm^{-1}) show a well-defined peak formed by the unattenuated photons at the focal point of the beam and a uniformly fluctuating background around the peak. Because of the photon of the incident beam are treated as a spherical wave in our Monte Carlo simulations, the unattenuated photons propagates toward the z-axis after emerging from the slab and form a peak at the focal point. This peak in photon density has a significant width along the z-axis because of the aberrations of the wavefront suffered at the plane interfaces of the slab while appear as a single point at the focal point in the xy-plane. Figs. 2 and 3 show the light distribution in the xy-plane and xz-plane corresponding to the case of $\mu_t = 0.936(mm^{-1})$. To obtain these results, a total of 1.14×10^9 photons are injected at the entrance surface of the slab phantom to produce a photon density in background and peak with negligible statistical fluctuations. Considering the reflection at the entrance surface, a total of 1.115×10^9 photons enter the slab and are tracked by the program. Twenty tasks are used in the parallel calculation, and it takes the PC cluster about 9 hours to finish.

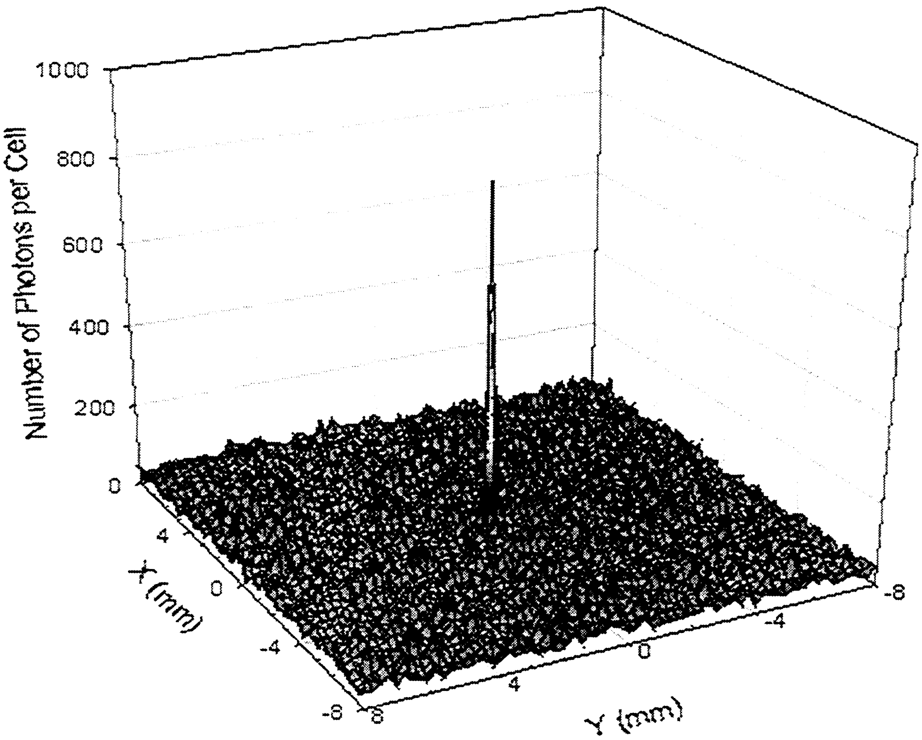

Fig. 3 Same as Fig. 2 except in the xy-plane at the focal point $z = Z_f = 67.3mm$.

Fig. 4 shows the dependence of the number of unattenuated photons arrived at the focal point on the attenuation coefficient μ_t when μ_t is changed from 0.546 (mm^{-1}) to 1.092 (mm^{-1}). The unattenuated photons at the focal point is obtained by subtracting the background formed by the diffusely scattered photons from the observed peak. To do that, we first obtain the photon density, or the number of photons per cell at the focal point, N_f, from the light distribution deposition in the yz-plane, and N_f includes both unattenuated and diffusely scattered photons. The diffusely scattered photon density N_b, i.e. the background in the photon density near the focal point, is obtained by averaging the number of photons per cell over the cells in the xy-plane excluding the single cell where the peak is located. A simple treatment like this is well justified. First: the deposit region is small enough so that the background is relatively flat. Second: due to the geometric optical approximation adopted in the Monte Carlo simulation, all the unattenuated photons are focused to a single cell in the xy-plane at $z = Z_f$, so by excluding that single cell, the xy-plane distribution is dominated by that of the diffusely scattered photons near the focal point. In this way we can obtain the unattenuated photons at the focal point by $N_{unatt} = (N_f - N_b)$ for each slab of μ_t. The results shown in Fig. 4 (solid circle) are in excellent agreement with the radiative transfer theory prediction (the solid line) given by Eq. (8). This proves directly that the photon density peak at the focal point consists of the unattenuated photons on top of the diffusive background formed by the multiply scattered photons.

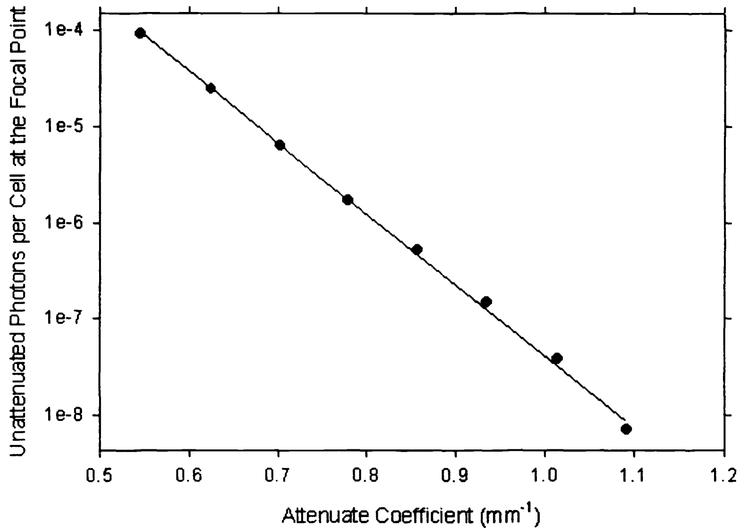

Fig. 4 Parallel computing results of Monte Carlo simulation of the dependency of unattenuated photons near the focal point on the attenuate coefficient μ_t. The solid circles are the simulation results, while the straight is the predicted value from Eq. (8).

To produce a background in the photon number per cell with negligible fluctuation, there is a minimum for the number of photons N_0 required for the incident beam even when μ_t is small. In our cases shown in Fig. 4 it is $N_0 = 3.5 \times 10^8$ for $\mu_t \leq 0.858 mm^{-1}$. And as μ_s increases, more photons are needed to reduce statistical fluctuation for the peak at the focal point. For example, the number N_0 is increased to 1.14×10^9 for $\mu_t = 0.936 mm^{-1}$, 2.2×10^9 for $\mu_t = 1.014 mm^{-1}$, and 5.6×10^9 for $\mu_t = 1.092 mm^{-1}$ for the results shown in Fig. 4. The steep increase in the total number of tracked photon as a result of large optical depth clearly demonstrates the urgent need to adopt the parallel computing technique for conducting high performance computation in the biomedical optics.

In summary, we have adopted the parallel computing technique in our simulation of light propagating in highly turbid media of biological tissues. Our results have shown that the parallel technique can be implemented into Monte Carlo simulations to calculate light distribution with inhomogeneous boundary conditions. With nearly no overhead cost in the CPU time for distributed parallel computing and the rapid increase in the performance/cost ratio of PC, this approach offers excellent opportunity in the near future to conduct large-scale Monte Carlo simulations for the modeling of light-tissue interaction. For example, the expansion of the current work to include surface roughness is in progress and will be reported elsewhere.

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